

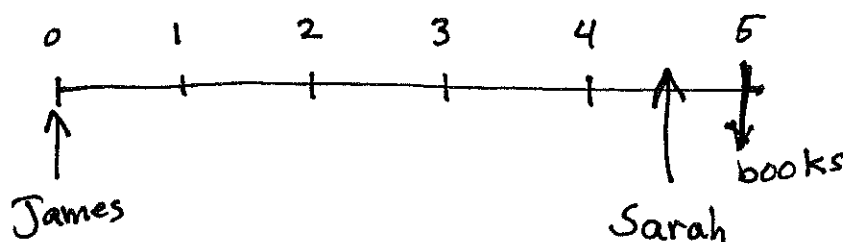
PHYS1408-2 Exam 3 November 20, 2015

$$\rho_{\text{air}} = 1.21 \frac{\text{kg}}{\text{m}^3} \quad \rho_{\text{ice}} = 917 \frac{\text{kg}}{\text{m}^3} \quad \rho_{\text{water}} = 1000 \frac{\text{kg}}{\text{m}^3} \quad \rho_{\text{seawater}} = 1028 \frac{\text{kg}}{\text{m}^3}$$

$$P_0 = 1.013 \times 10^5 \frac{\text{N}}{\text{m}^2} \quad G = 6.67 \times 10^{-11} \frac{\text{N m}^2}{\text{kg}^2}$$

$$M_{\text{earth}} = 5.98 \times 10^{24} \text{ kg} \quad R_{\text{earth}} = 6.37 \times 10^6 \text{ m}$$

1. A 100 kg sofa 5 m long is carried by James and Sarah. James has the end, Sarah is holding the sofa 0.5 m from the other end. A passing helper places a 50 kg box of books on the sofa on the very end of the sofa Sarah is holding.



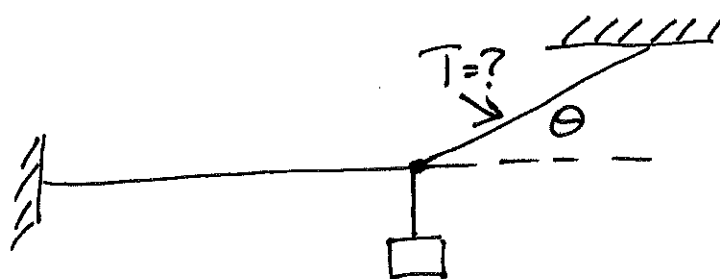
How much weight is Sarah holding now?

a) 380 N b) 490 N c) 980 N d) 1090 N e) 1470 N

2. A 150 kg man (and the load he carries) is going up a leaning 20 kg ladder. The ladder is set at 71° . The floor and ladder have a coefficient of friction of 0.25. The wall is frictionless. How far up the 5 m long ladder can he safely climb?

a) 3.0 m b) 3.5 m c) 4.0 m d) 4.5 m e) 5.0 m

3. A light (20 kg) is attached to the ceiling by a cord making an angle 40° with the ceiling. The second cord is exactly horizontal and anchored in the wall. What is the tension in the cord connected to the ceiling?



a) 200 N b) 300 N c) 400 N d) 500 N e) 600 N

4. A block of jello $Y = 3.00 \times 10^4 \text{ N/m}^2$ is 3 m wide, 4 m long, and 0.1 m high is compressed by a mass of 500 kg. By how much does the height shrink?

a) 0.3 mm b) 0.5 mm c) 0.9 mm d) 1.1 mm e) 1.3 mm

5. A load of 500 kg is hung from a circular wire 4.00 m long, diameter 0.001 m, and a Young's modulus of $8.00 \times 10^{10} \text{ N/m}^2$. By how much does the wire stretch?

a) 0.003 m b) 0.009 m c) 0.03 m d) 0.09 m e) 0.3 m

6. A chunk of ice floats in the sea. What fraction of the ice is below the waterline?

- a) 88% b) 89% c) 90% d) 91% e) 92%

7. A 3.0 kg block of metal is massed on a spring scale in air. When the block is hung on the spring scale but the block suspended in water the scale reads 0.5 kg. What is the density of the metal?

- a) 1200 kg/m³ b) 1400 kg/m³ c) 1600 kg/m³ d) 1800 kg/m³ e) 2000 kg/m³

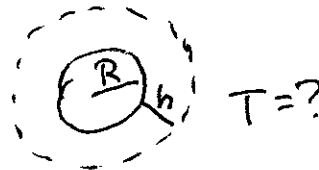
8. Water flows out of a large open reservoir through a small pipe that opens to the atmosphere 30 m below the level of the water in the reservoir. What is the speed of the water as it comes out of the pipe?

- a) 4 m/s b) 9 m/s c) 14 m/s d) 19 m/s e) 24 m/s

9. A rocket fired from earth's surface rises 1000 km. At what speed was it launched?

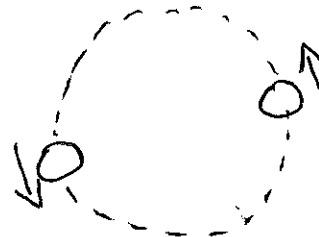
- a) 640 m/s b) 900 m/s c) 3 km/s d) 8 km/s e) 11 km/s

10. What is the period of a satellite (mass 1000 kg) in circular orbit around the earth at an altitude of 10^8 m above the earth's surface?



- a) 5 minutes b) 1 hour c) 10 hours d) 4 days e) 40 days

11. What is the period of the circular orbit of two equal mass stars (10^{31} kg each) around their common center of mass (halfway between them)?



Separation distance $d = 10^8$ m

- a) 3 minutes b) 30 minutes c) 5 hours d) 2 days e) 20 days

12. What is the acceleration due to gravity on a 120 kg man at a height of 750 km above the Earth's surface?

- a) 7 m/s² b) 8 m/s² c) 9 m/s² d) 10 m/s² e) 700 m/s²

13. The earth and the moon (mass 7×10^{22} kg) are separated by 400,000 km, from center to center. How far from the center of the earth toward the moon would you have move to find a place where the gravitational pull towards the earth would equal the gravitational pull towards the moon?

- a) 50% b) 90% c) 96% d) 99% e) 99.8%